ESP-IDF ESP32-S3 MAPA OFFLINE (MENUCONFIG)

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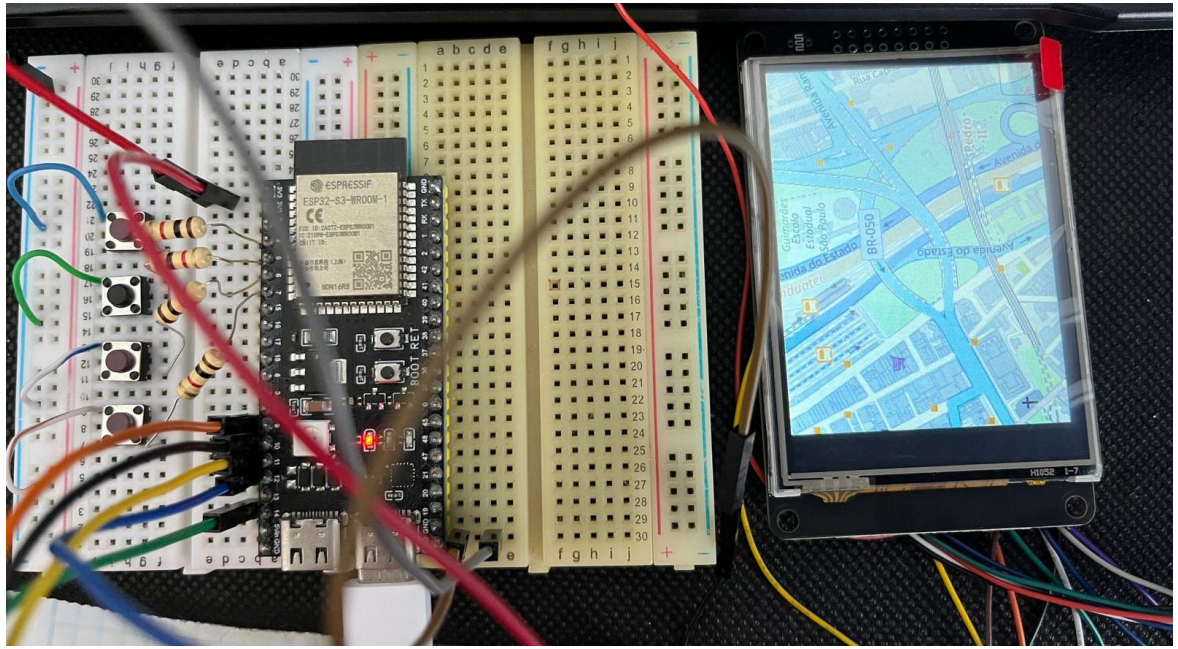
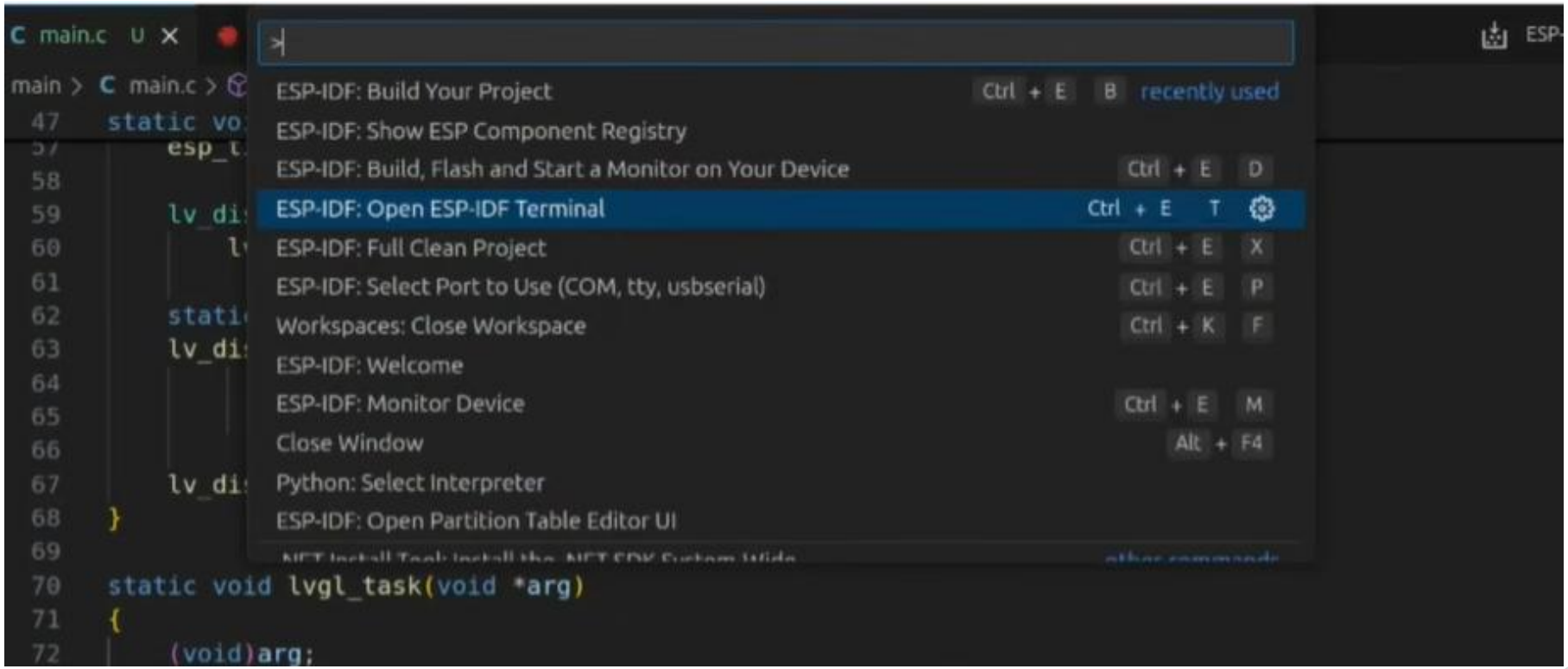


https://github.com/alcarazolabs/mapa-offline-esp32s3-display-240x320-st7789

1. Abrir ESP-IDF Terminal



2. Ejecutar “idf.py menuconfig”

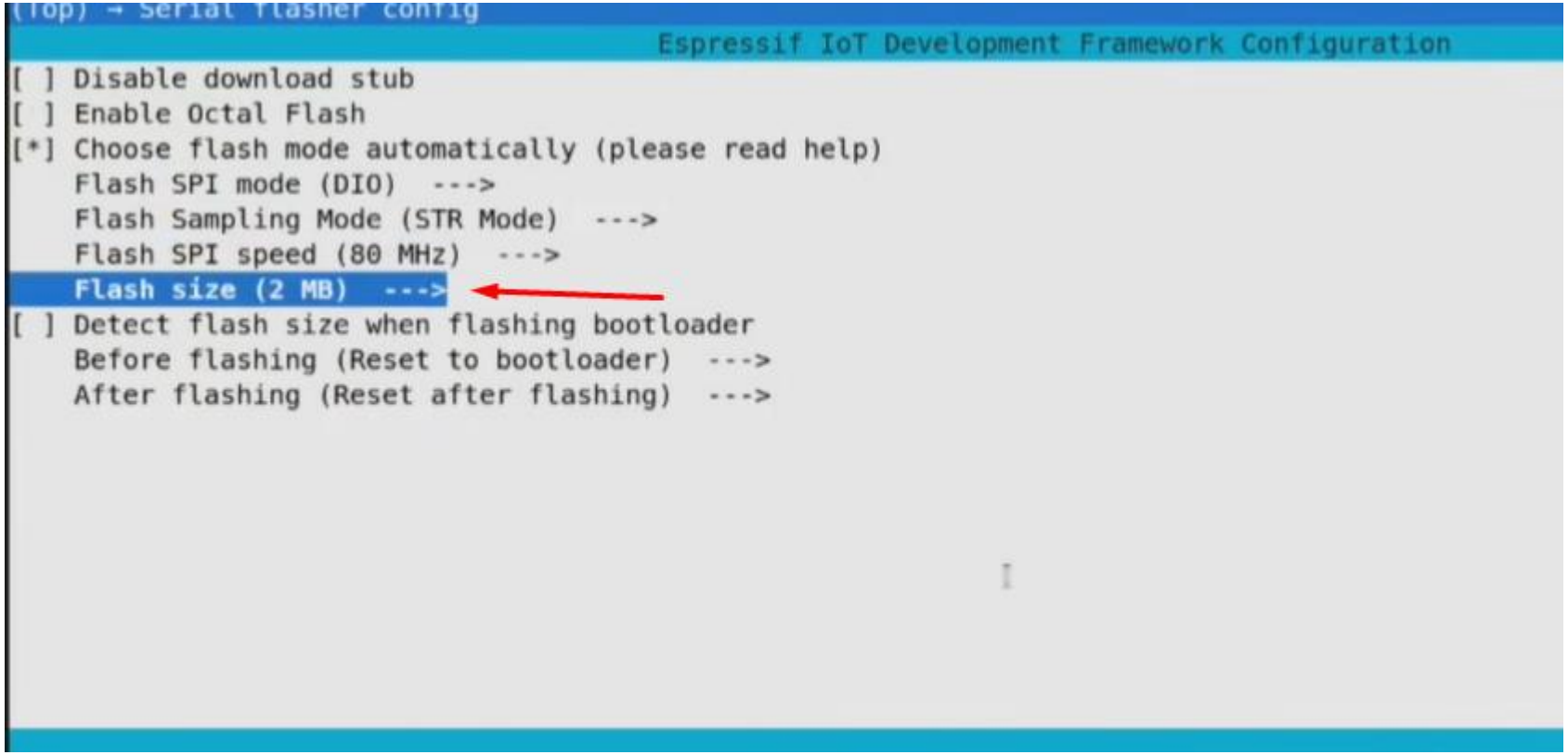
```
65                                  LV_DISPLAY_RENDER_MODE_PARTIAL);
66
67          lv_display_set_flush_cb(display, lvgl_flush_cb);
68      }
69
70      static void lvgl_task(void *arg)
71      {
```

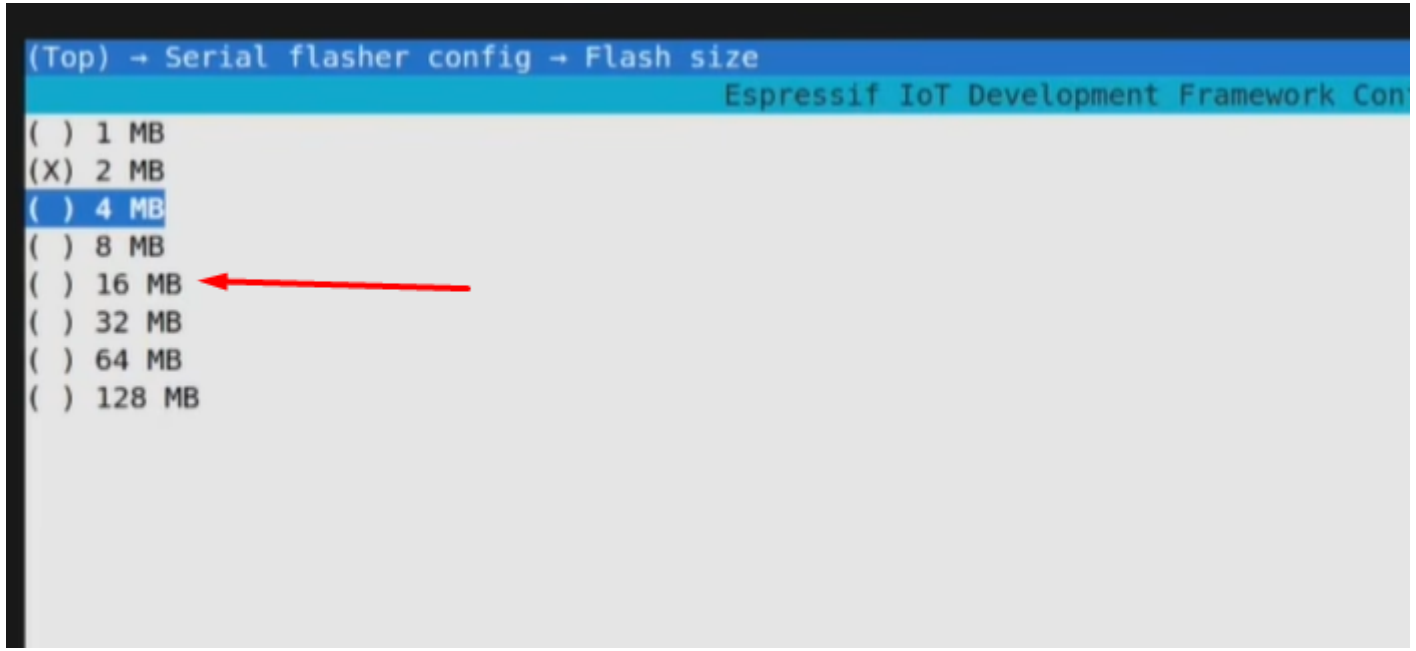
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS ESP-IDF SERIAL MONITOR COMMENTS ESP-IDF Termin

```
gustavo@gustavo-730QED:~/repo/map-viewer-offline-esp32s3$ idf.py menuconfig
```

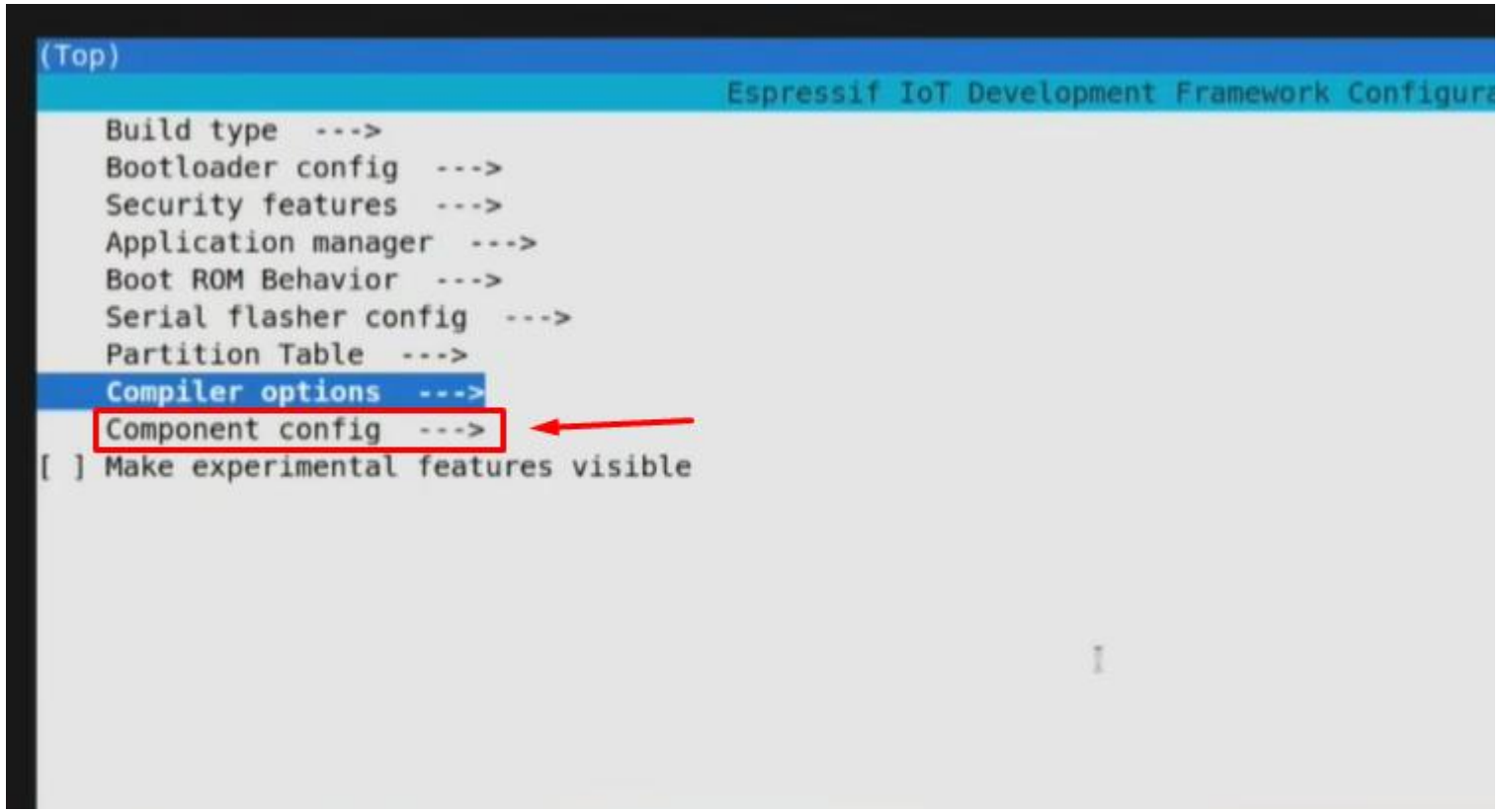
Configurar memoria flash

```
(Top)
                    Espressif IoT Development Framework Configuration
    Build type  --->
    Bootloader config  --->
    Security features  --->
    Application manager  --->
    Boot ROM Behavior  --->
    Serial flasher config  --->
    Partition Table  --->
    Compiler options  --->
    Component config  --->
[ ] Make experimental features visible
```


(Top) → Serial flasher config
Espressif IoT Development Framework Configuration
[ ] Disable download stub
[ ] Enable Octal Flash
[*] Choose flash mode automatically (please read help)
Flash SPI mode (DIO) --->
Flash Sampling Mode (STR Mode) --->
Flash SPI speed (80 MHz) --->
Flash size (2 MB) --->
[ ] Detect flash size when flashing bootloader
Before flashing (Reset to bootloader) --->
After flashing (Reset after flashing) --->


(Top) → Serial flasher config → Flash size
Espressif IoT Development Framework Con
( ) 1 MB
(X) 2 MB
( ) 4 MB
( ) 8 MB
( ) 16 MB
( ) 32 MB
( ) 64 MB
( ) 128 MB

Configurar PSRAM



```
(Top) → Component config
    ↑↑↑↑↑↑↑↑↑↑↑↑↑↑                    Espressif IoT Development Framework Configuration
    Ethernet  --->
    Event Loop Library  --->
    GDB Stub  --->
    ESP HID  --->
    ESP HTTP client  --->
    HTTP Server  --->
    ESP HTTPS OTA  --->
    ESP HTTPS server  --->
    Hardware Settings  --->
    ESP-Driver:LCD Controller Configurations  --->
    ESP-MM: Memory Management Configurations  --->
    ESP NETIF Adapter  --->
    Partition API Configuration  ----
    PHY  --->
    Power Management  --->
    ESP PSRAM  --->
    ESP Ringbuf  --->
    ESP-ROM  ----
    ESP Security Specific  ----
    ↓↓↓↓↓↓↓↓↓↓↓↓↓↓
[Space/Enter] Toggle/enter  [ESC] Leave menu          [S] Save
[O] Load                    [?] Symbol info           [/] Jump to symbol
[F] Toggle show-help mode   [C] Toggle show-name mode [A] Toggle show-all mode
```

```
(Top) → Component config → ESP PSRAM
                    Espressif IoT Development Framework Configuration
[ ] Support for external, SPI-connected RAM
```

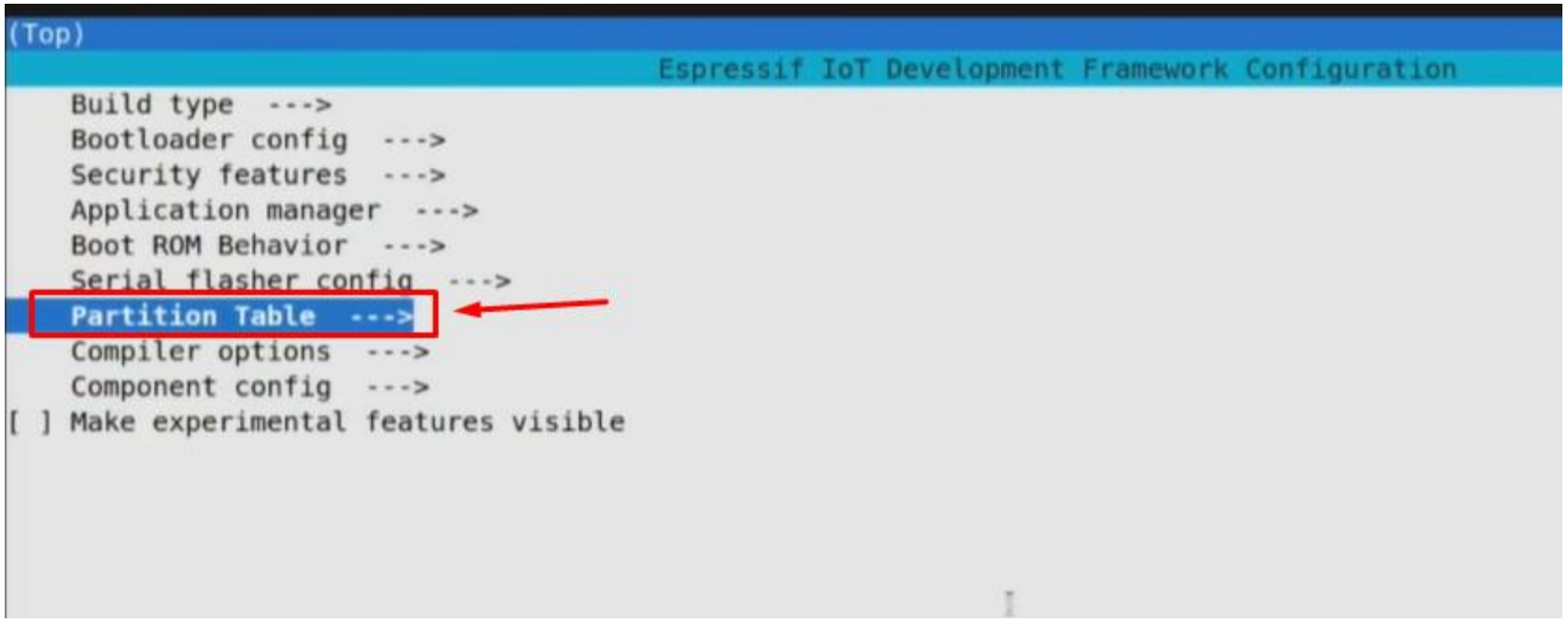
```
(Top) → Component config → ESP PSRAM → Support for external, SPI-connected RAM → SPI RAM config
                    Espressif IoT Development Framework Configuration
    Mode (QUAD/OCT) of SPI RAM chip in use (Quad Mode PSRAM)  --->
    Type of SPIRAM chip in use (Auto-detect)  --->
[ ] Enable Executable in place from (XiP) from PSRAM feature (NEW)
[ ] Move Instructions in Flash to PSRAM (NEW)
[ ] Move Read-Only Data in Flash to PSRAM (NEW)
    Set RAM clock speed (40Mhz clock speed)  --->
[*] Initialise the PSRAM related hardware (NEW)
[*]     Initialize PSRAM during startup, including the hardware and memory related configurations (NEW)
[*] Pre-configure memory protection for PSRAM (NEW)
[ ] Ignore PSRAM when not found (NEW)
    SPI RAM access method (Make RAM allocatable using malloc() as well)  --->
[*] Run memory test on SPI RAM initialization (NEW)
(16384) Maximum malloc() size, in bytes, to always put in internal memory (NEW)
[ ] Try to allocate memories of WiFi and LWIP in SPIRAM firstly. If failed, allocate internal memory (NEW)
(32768) Reserve this amount of bytes for data that specifically needs to be in DMA or internal memory (NEW)
[ ] Allow .bss segment placed in external memory (NEW)
[ ] Allow .noinit segment placed in external memory (NEW)
```

```
Component config → ESP PSRAM → Support for external, SPI-connected RAM → SPI RAM config → Mode (QUAD/OCT) of SPI RAM chip in u
                    Espressif IoT Development Framework Configuration
(X) Quad Mode PSRAM
( ) Octal Mode PSRAM
```

```
(Top) → Component config → ESP PSRAM → Support for external, SPI-connected RAM → SPI RAM config
                                Espressif IoT Development Framework Configuration
    Mode (QUAD/OCT) of SPI RAM chip in use (Octal Mode PSRAM)  --->
    Type of SPIRAM chip in use (Auto-detect)  --->
[ ] Enable Executable in place from (XiP) from PSRAM feature (NEW)
[ ] Move Instructions in Flash to PSRAM (NEW)
[ ] Move Read-Only Data in Flash to PSRAM (NEW)
    Set RAM clock speed (40Mhz clock speed)  --->
[ ] Enable SPI RAM ECC (NEW)
[*] Initialise the PSRAM related hardware (NEW)
[*]     Initialize PSRAM during startup, including the hardware and memory related configurations (NEW)
[*] Pre-configure memory protection for PSRAM (NEW)
[ ] Ignore PSRAM when not found (NEW)
    SPI RAM access method (Make RAM allocatable using malloc() as well)  --->
[*] Run memory test on SPI RAM initialization (NEW)
(16384) Maximum malloc() size, in bytes, to always put in internal memory (NEW)
[ ] Try to allocate memories of WiFi and LWIP in SPIRAM firstly. If failed, allocate internal memory (NEW)
(32768) Reserve this amount of bytes for data that specifically needs to be in DMA or internal memory (NEW)
[ ] Allow .bss segment placed in external memory (NEW)
[ ] Allow .noinit segment placed in external memory (NEW)

[Space/Enter] Toggle/enter  [ESC] Leave menu           [S] Save
[O] Load                    [?] Symbol info            [/] Jump to symbol
[F] Toggle show-help mode   [C] Toggle show-name mode  [A] Toggle show-all mode
[Q] Quit (prompts for save) [D] Save minimal config (advanced)
```

Realiza configuración de tabla de particion



```
(Top) → Partition Table
                    Espressif IoT Development Framework Configuration
    Partition Table (Single factory app, no OTA)  --->
(0x8000) Offset of partition table
[*] Generate an MD5 checksum for the partition table
```

```
(Top) → Partition Table → Partition Table
                    Espressif IoT Development Framework Configu
(X) Single factory app, no OTA
( ) Single factory app (large), no OTA
( ) Factory app, two OTA definitions
( ) Two large size OTA partitions
( ) Custom partition table CSV
```

Luego regresar al menú principal:

```
(Top)
                    Espressif IoT Development Framework Configuration
    Build type  --->
    Bootloader config  --->
    Security features  --->
    Application manager  --->
    Boot ROM Behavior  --->
    Serial flasher config  --->
    Partition Table  --->
    Compiler options  --->
    Component config  --->
[ ] Make experimental features visible
```

Presionar “S” para guardar y “Q” para salir.